

Equivalence Completion Between Yukawa and Mass Bookkeeping

Amos Jay Maley

Abstract

We complete an equivalence map between electroweak Higgs–Yukawa bookkeeping and low-energy mass-based bookkeeping. Parameters commonly treated as independent in ultraviolet descriptions are shown to overcount tensor load once admissible redescription and regime transport are enforced. The fermion sector admits a single invariant witness per species. The result is eliminative, invariant under admissible redescription, and introduces no new physics.

1 Apparent Independence in Higgs–Yukawa Bookkeeping

In electroweak bookkeeping with a scalar doublet, fermion masses arise from Yukawa interactions

$$\mathcal{L}_Y = -y_f \bar{\psi}_{f,L} \phi \psi_{f,R} + \text{h.c.} \quad (1)$$

After symmetry breaking, the fermion mass is

$$m_f = \frac{y_f v}{\sqrt{2}}, \quad (2)$$

where y_f and v are treated as independent inputs in this skin.

2 Admissible Low-Energy Bookkeeping

An admissible low-energy description of the same fermion sector contains massive fermion fields with mass parameter m_f . Interaction strengths, thresholds, and propagators depend on m_f and gauge couplings. No independent parameter corresponding to y_f appears.

3 Lemma 1: Cross-Skin Transport

Admissible transport from the Higgs–Yukawa skin to the low-energy massive-fermion skin identifies

$$m_f \leftrightarrow \frac{y_f v}{\sqrt{2}}. \quad (3)$$

All fermion-sector amplitudes in the low-energy skin depend on m_f alone.

4 Lemma 2: Invariant Witness Counting

If a tensor load contains n independent degrees of freedom, admissible redescription and regime transport must preserve n independent invariant witnesses across all admissible skins describing the same regime. A skin exposing fewer invariant witnesses implies redundancy in higher-dimensional parameterizations.

5 Failure of Independence

Assume (y_f, v) are independent tensor degrees for fermion threshold and interaction content. By Lemma 2, two invariant witnesses must survive admissible transport. The low-energy skin exposes only one invariant witness, m_f . Preservation of independence would require loss of tensor load, hidden unwitnessed structure, or representational privilege. Each constitutes standing failure.

6 Forced Tensor Relation

The admissible fermion-sector tensor load is uniquely specified by

$$\boxed{m_f \equiv \frac{y_f v}{\sqrt{2}}} \tag{4}$$

for each fermion species f . The Yukawa parameter y_f is not independently admissible as a tensor degree for fermion threshold or interaction content once m_f is fixed.

7 Corollaries

- The fermion sector admits a one-dimensional tensor load per species.
- Higgs–Yukawa and mass-based descriptions are tensor-identical after redundancy elimination.
- Any framework treating (y_f, v) as independently physical for low-energy fermion content overcounts tensor degrees and violates standing.

Conclusion

Equivalence completion enforces collapse of apparent freedom between Yukawa and mass book-keeping. The result is eliminative, representation-invariant, and irreversible under admissible re-description.